

Amplitude Damping Channel

$$|0\rangle_s \otimes |0\rangle_e \rightarrow |0\rangle_s \otimes |0\rangle_e$$

$$|1\rangle_s \otimes |0\rangle_e \rightarrow \sqrt{1-p} |1\rangle_s \otimes |0\rangle_e + \sqrt{p} |0\rangle_s \otimes |1\rangle_e$$

$$M_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad M_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

$$\Lambda_p(\rho) = \sum_a M_a \rho M_a^\dagger = \begin{pmatrix} \rho_{00} + p \rho_{11} & \sqrt{1-p} \rho_{01} \\ \sqrt{1-p} \rho_{10} & (1-p) \rho_{11} \end{pmatrix}$$

